

ASSESSMENT : CompTia A+ - Troubleshooting and Helpdesk
Section B — Practical Coding Tasks

PRACTICAL TASK 1

1. Identify and describe the function of the motherboard, CPU, RAM, GPU, storage device, and cooling system.
2. Note the make/model or key specification of each component where visible.
3. Present the findings as a labeled component list or table.

Component Identification & Functions

System Used: ASUS TUF Gaming F15 FX506HC

Practical Method: Physical inspection of laptop hardware and Windows system information.

Component Identification Table

Component	Make / Model / Specification	Function
Motherboard	ASUS TUF Gaming F15 FX506HC motherboard	The motherboard is the main circuit board of the system. It connects the CPU, RAM, storage, GPU, cooling system, battery and other hardware and allows them to communicate with each other.
CPU	11th Gen Intel® Core™ i5-11400H @ 2.70 GHz	The CPU executes instructions and performs general computing tasks required by the operating system and applications.
RAM	16 GB, 3200 MT/s; Crucial RAM module visible	RAM provides temporary high-speed memory for data and instructions that are currently being used by the operating system and applications.
GPU	NVIDIA GeForce RTX 3050 Laptop GPU, 4 GB	The GPU processes graphics and visual data. It supports graphics-intensive applications, games and GPU-accelerated workloads.
Integrated GPU	Intel® UHD Graphics	Provides integrated graphics processing for supported display and graphics workloads.

Storage	Internal storage, 477 GB capacity shown in Windows	Storage permanently holds the operating system, applications, documents and other user data.
Cooling System	Dual cooling fans with copper heat pipes and heatsinks	The cooling system removes heat generated by the CPU and GPU and helps maintain safe temperatures during prolonged workloads.

Physical Identification

The laptop was opened for physical inspection. The internal photograph clearly shows the **ASUS motherboard, two cooling fans, copper heat pipes/heatsinks, RAM modules, storage area and internal battery**.

The RAM modules are installed on the motherboard and one module has a visible **Crucial** label. The cooling system consists of two blower-style fans connected to copper heat pipes and heatsinks.

The CPU and GPU are located underneath the cooling assembly, so their physical chips are not directly visible. Their exact models were confirmed using **Windows → Settings → System → About**.

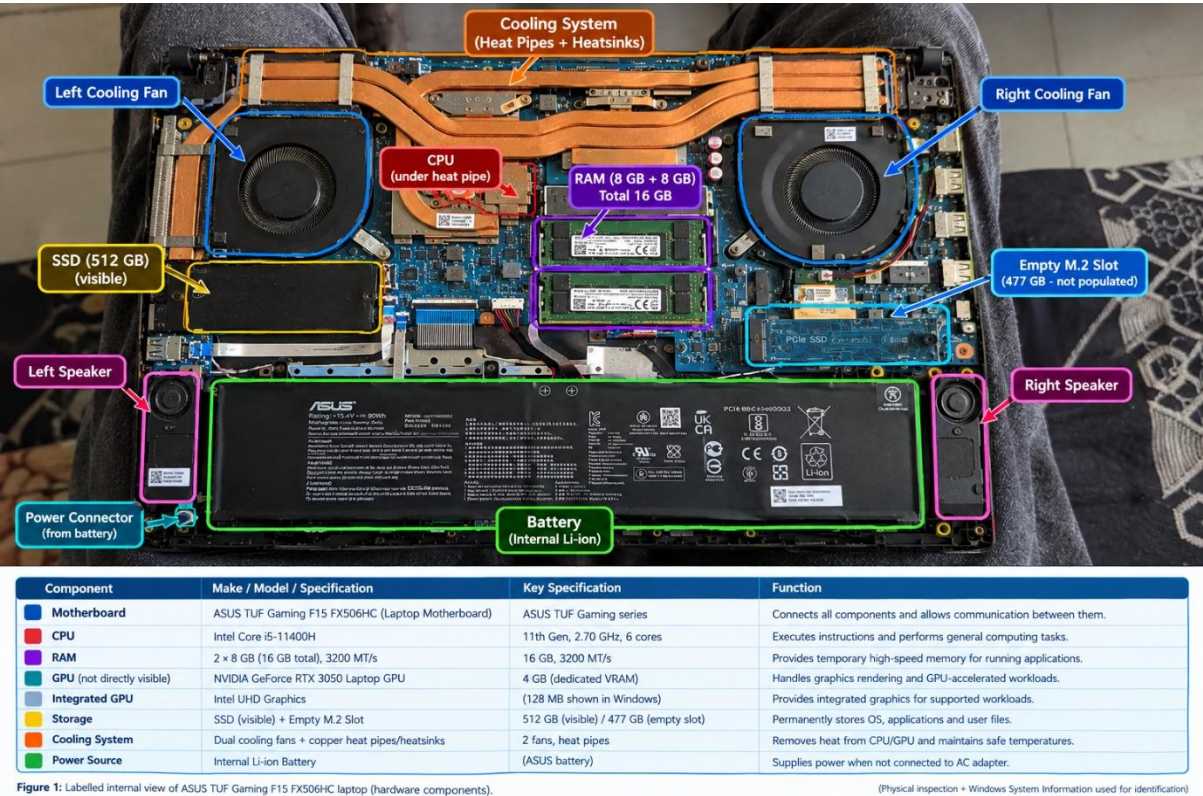
System Information

Windows system information confirmed:

- **Processor:** 11th Gen Intel Core i5-11400H @ 2.70 GHz
- **Installed RAM:** 16 GB
- **RAM Speed:** 3200 MT/s
- **Graphics:** NVIDIA GeForce RTX 3050 Laptop GPU — 4 GB
- **Integrated Graphics:** Intel UHD Graphics
- **Storage:** 477 GB shown by Windows
- **System:** 64-bit operating system, x64-based processor

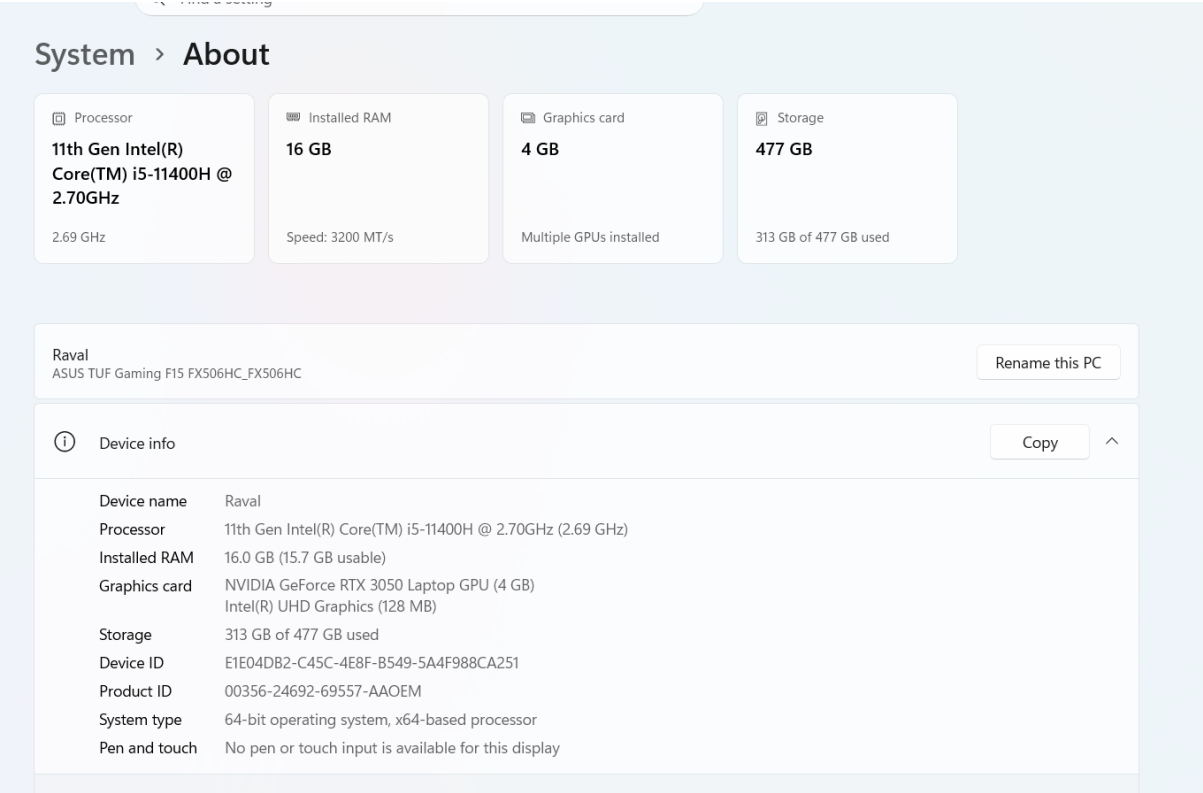
Screenshot:

Figure 1 — Physical Hardware:



Internal photograph showing the laptop motherboard, RAM, cooling fans, copper heat pipes, storage area and battery.

Figure 2 — System Information:



Windows Settings → System → About showing the processor, installed RAM, graphics hardware and storage information.

Conclusion

- The practical inspection successfully identified the main hardware components of the ASUS TUF Gaming F15 workstation.
- The physical inspection was combined with Windows system information to verify the available specifications.
- Each component has a specific role, and together they provide the processing, memory, graphics, storage and thermal management required for normal workstation operation.

PRACTICAL TASK 2

Motherboard Ports & Socket Identification

Question:

Identify the motherboard's physical layout, ports, and CPU socket type.

Practical Requirements:

- Identify and label all major motherboard ports (SATA, power, USB headers, etc.).
- Identify expansion slots (PCIe, M.2, etc.) and their purpose.
- Identify the CPU socket type and note which CPU families are compatible with it.
- Photograph or diagram the motherboard with the components labelled.

Answer / Practical Documentation:

The motherboard inspected was from an **ASUS TUF Gaming F15 FX506HC** laptop. The internal motherboard layout was examined with the bottom cover removed.

1. Motherboard Layout

The motherboard is the main printed circuit board (PCB) of the laptop. It contains the processor, memory slots, storage connections, power circuits, cooling connections and external I/O connections.

The following components were identified during the physical inspection:

Component / Port	Identification	Purpose
Motherboard	ASUS TUF Gaming F15 FX506HC	Connects and allows communication between the system components.
RAM slots	2 × SO-DIMM slots	Hold the system RAM modules.
RAM installed	2 × 8 GB = 16 GB	Provides temporary working memory for the operating system and applications.
M.2 PCIe slot	One populated M.2 SSD area	Connects the NVMe/PCIe SSD to the motherboard.
Additional M.2 slot	Empty M.2 slot	Provides an additional location for compatible PCIe/NVMe storage.
Cooling fan connectors	Fan connections near the cooling assemblies	Supply power/control signals to the cooling fans.

Battery connector	Internal battery connection	Connects the battery to the motherboard's power circuitry.
Display/other internal connectors	Ribbon and board connectors visible on the motherboard	Connect internal laptop components and peripherals to the motherboard.

2. Expansion and Storage Slots

The laptop contains **M.2 PCIe storage connectivity**. One M.2 slot is populated with a **512 GB SSD**, while another M.2 slot is currently empty.

The M.2 PCIe interface allows a compatible NVMe SSD to communicate with the motherboard using PCIe lanes, providing high-speed storage performance.

The laptop does not use a conventional desktop PCIe expansion slot for installing a full-size desktop graphics card. The **NVIDIA GeForce RTX 3050 Laptop GPU** is integrated/soldered onto the laptop motherboard and is not a removable PCIe graphics card.

3. CPU Socket Identification

The **Intel Core i5-11400H** processor is used in this laptop. Unlike a typical desktop processor, the laptop CPU is **soldered directly to the motherboard** rather than being installed in a user-replaceable socket.

Therefore, there is **no conventional removable CPU socket visible** for this processor in the physical inspection. The CPU is located underneath the copper heatsink/heat-pipe assembly.

The Intel Core i5-11400H belongs to Intel's **11th Generation mobile H-series** processors and uses a **BGA package**. This means the processor is soldered to the motherboard and is not designed to be removed and replaced like a desktop LGA processor.

4. External Motherboard I/O

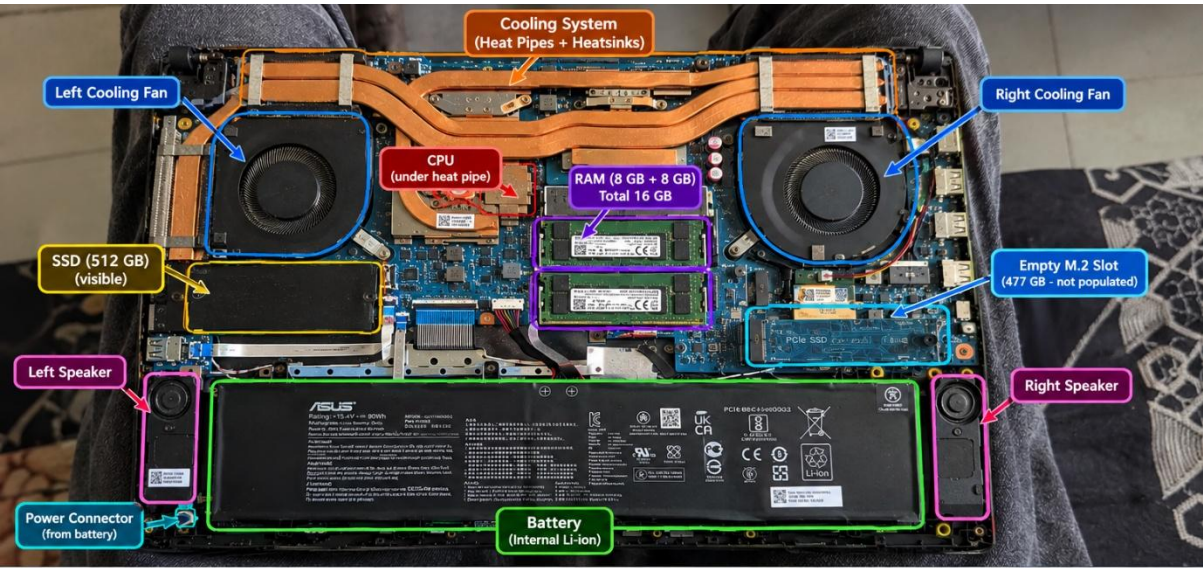
The laptop's motherboard provides connections for external devices through the side I/O ports. These include USB, display/video, networking and audio-related interfaces appropriate to the laptop model.

These ports allow external peripherals such as keyboards, mice, storage devices, displays and network equipment to be connected.

Conclusion

The physical inspection identified the ASUS motherboard, two RAM slots, M.2 PCIe storage connections, cooling connections and internal power connections. The system contains **2 × 8 GB RAM (16 GB total)** and a **512 GB SSD**, with an additional M.2 slot available for compatible storage. The Intel Core i5-11400H is a soldered BGA laptop processor, so it does not have a conventional user-replaceable CPU socket.

Screenshot:



Component	Make / Model / Specification	Key Specification	Function
Motherboard	ASUS TUF Gaming F15 FX506HC (Laptop Motherboard)	ASUS TUF Gaming series	Connects all components and allows communication between them.
CPU	Intel Core i5-11400H	11th Gen, 2.70 GHz, 6 cores	Executes instructions and performs general computing tasks.
RAM	2 × 8 GB (16 GB total), 3200 MT/s	16 GB, 3200 MT/s	Provides temporary high-speed memory for running applications.
GPU (not directly visible)	NVIDIA GeForce RTX 3050 Laptop GPU	4 GB (dedicated VRAM)	Handles graphics rendering and GPU-accelerated workloads.
Integrated GPU	Intel UHD Graphics	(128 MB shown in Windows)	Provides integrated graphics for supported workloads.
Storage	SSD (visible) + Empty M.2 Slot	512 GB (visible) / 477 GB (empty slot)	Permanently stores OS, applications and user files.
Cooling System	Dual cooling fans + copper heat pipes/heatsinks	2 fans, heat pipes	Removes heat from CPU/GPU and maintains safe temperatures.
Power Source	Internal Li-ion Battery	(ASUS battery)	Supplies power when not connected to AC adapter.

Figure 1: Labelled internal view of ASUS TUF Gaming F15 FX506HC laptop (hardware components). (Physical inspection + Windows System Information used for identification)

PRACTICAL TASK 3

RAM & Storage Installation

Question:

Remove and reinstall a RAM module and a storage device, following safe hardware handling procedures. Confirm that both components are detected correctly after reinstallation.

Practical Work / Answer:

System Used: ASUS TUF Gaming F15 FX506HC

Components handled:

- RAM: **2 × 8 GB DDR4, 3200 MT/s (16 GB total)**
- Storage: **512 GB M.2 PCIe SSD**

1. Safety Precautions

Before working on the internal components, I shut down the laptop completely and disconnected the AC adapter. I ensured that the laptop was not connected to external power and handled the internal components carefully to avoid electrostatic discharge (ESD).

I also avoided touching the gold contacts of the RAM and SSD and held the components by their edges.

2. RAM Removal and Reinstallation

1. The laptop's bottom cover was removed carefully.
2. The internal battery connection was identified before working on the components.
3. The retaining clips on the sides of the RAM module were released.
4. The RAM module was lifted carefully from the SO-DIMM slot.
5. The module was inspected for damage or contamination.
6. The RAM module was aligned with the notch in the SO-DIMM slot.
7. It was inserted at an angle and then pressed down until the retaining clips locked it into position.
8. The same procedure was followed for the second RAM module if required.

The laptop contains **two 8 GB RAM modules**, giving a total installed capacity of **16 GB at 3200 MT/s**.

3. Storage Removal and Reinstallation

1. The M.2 SSD was located on the motherboard.
2. The retaining screw was removed carefully.
3. The M.2 SSD was lifted slightly and removed from the slot.
4. The SSD and connector were inspected for visible damage.
5. The SSD was reinserted into the M.2 PCIe slot at the correct angle.
6. It was pressed down and secured using the retaining screw.
7. The connection was checked to ensure that the SSD was seated correctly.

The installed storage device is a **512 GB M.2 PCIe SSD**.

4. Post-Installation Verification

After reinstalling the components, the laptop was powered on and the hardware was checked.

In Windows, **Settings → System → About** confirmed:

- **Installed RAM:** 16 GB
- **RAM Speed:** 3200 MT/s
- **Storage:** approximately 477 GB usable capacity shown by Windows

The storage capacity displayed by Windows is lower than the manufacturer's 512 GB rating because manufacturers calculate capacity using decimal units, while Windows reports storage capacity differently and some space is also used for system/partition information.

Conclusion

The RAM and M.2 PCIe SSD were handled using appropriate safety precautions and reinstalled correctly. After reinstallation, Windows successfully detected the **16 GB RAM** and the **512 GB-class SSD**, confirming that both components were functioning correctly.

PRACTICAL TASK 4

Full Desktop PC Assembly

Question:

Assemble a desktop PC from the provided components and document the process. Include correct installation sequence and ESD/safety precautions. Document each step with labelled photos or diagrams. Power on the system and confirm it passes POST successfully.

Practical Method:

Physical desktop hardware was not available. Therefore, the assembly procedure was documented using a clearly labelled hardware assembly sequence/diagram, following standard PC-building procedures.

1. Safety and Preparation

Before assembly, I would prepare a clean, dry workspace and disconnect all power sources. I would use an anti-static wrist strap or regularly ground myself to reduce the risk of electrostatic discharge (ESD).

I would inspect all components for physical damage and keep screws and small components organised.

2. Install the CPU

First, I would install the compatible CPU into the motherboard's CPU socket according to the manufacturer's instructions. I would ensure that the CPU alignment marker matches the socket marker and avoid touching the CPU contacts.

3. Install RAM

I would install the RAM modules into the appropriate motherboard DIMM slots. The notch on each RAM module must align with the slot. I would press the module down evenly until the retaining clips lock into place.

4. Install the CPU Cooler

I would install the CPU cooler using the correct mounting hardware. Thermal paste would be applied correctly if it is not already pre-applied. The CPU fan cable would then be connected to the appropriate motherboard fan header.

5. Install the Motherboard into the Case

I would install the correct motherboard standoffs in the case and carefully place the motherboard into position. I would secure it using the appropriate screws without overtightening them.

6. Install Storage

An M.2 NVMe SSD would be installed into the compatible M.2 slot and secured with its retaining screw. If using a SATA SSD or HDD, I would connect both the SATA data cable to the motherboard and SATA power cable from the PSU.

7. Install the PSU

I would mount the PSU securely in the case and route the required power cables. The PSU must have sufficient wattage for the CPU, GPU and other components.

8. Install the GPU

If a dedicated graphics card is required, I would install it into the appropriate PCIe x16 slot and secure it to the case. I would connect the required PCIe power cables from the PSU.

9. Connect Power and Other Cables

I would connect:

- 24-pin ATX motherboard power
- 8-pin or required CPU EPS power
- GPU power, if required
- SATA power, where applicable
- Front-panel power/reset connectors
- USB and audio front-panel connectors
- Case fan connectors

I would ensure every connector is correctly aligned and fully seated.

10. Final Inspection

Before powering on, I would check that:

- All components are seated correctly.
- RAM and GPU retaining clips are locked.
- CPU cooler is securely installed.
- Power connectors are fully inserted.
- No loose screws or metal objects are inside the case.
- No cables are obstructing the fans.

- The motherboard is correctly mounted on its standoffs.
- The PSU is switched off before the final connection to AC power.

11. First Boot and POST Verification

After completing the assembly and final safety inspection, I would connect the AC power and switch on the PSU. I would press the power button and observe the system during startup.

I would confirm that:

- The system powers on.
- CPU and case fans operate.
- The display shows the manufacturer's logo or BIOS/UEFI screen.
- RAM is detected.
- Storage is detected.
- No POST error codes, warning LEDs, or abnormal beep codes are present.

If the system reaches the BIOS/UEFI screen without hardware errors, the system has successfully completed POST.

Conclusion

- A desktop PC should be assembled in a systematic order, beginning with the CPU, RAM and cooler, followed by motherboard installation, storage, PSU, GPU and cable connections.
- ESD precautions and a final hardware inspection are essential before the first boot.
- Successful detection of the major hardware components and entry into BIOS/UEFI confirms that the system has passed POST.